

Multi-Agent AI Recruitment Assessment System

DOCUMENT 2: Complete Frontend User Guide & Step-by-Step Walkthrough

Introduction & Dual-Portal Architecture

The application provides a dual-portal experience tailored for both applicants and recruitment teams. Users can switch seamlessly at any time using the top navigation role switcher between the ■ **Candidate Portal** and the ■ **Recruiter Panel**.

Part 1: Candidate Portal (5-Step Lifecycle Guide)

Step 1: Upload PDF / Word Resume (Zero Typing)

- Drag and drop your .pdf, .docx, .doc, or .txt resume file into the upload zone.
- Alternatively, click any of the 1-click sample document buttons (Alex Chen Senior Backend, Jordan Miller FullStack AI, Elena Rostova Cloud Architect).
- The AI model parses the file, extracts technical competencies, work history, and education, and securely saves the candidate profile.

Step 2: Select Target Company Role & View Skill Gaps

- The portal displays matched open positions across top companies (Stripe, Anthropic, Scale AI, Snowflake, Databricks, OpenAI).
- Each company card displays the calculated **Role Compatibility %** (e.g. 94% Match).
- Click on a company card to review matching technical strengths (in green) and identified skill gaps (in amber).
- Click 'Proceed to Step 3: Take Coding Test' to begin the technical round.

Step 3: Live Technical Coding Assessment

- Read the algorithmic problem description (e.g. Two Sum with Hash Map optimization for O(N) performance).
- Write your Python solution directly in the embedded code editor.
- Click '**Execute & Submit Code**' to run the solution inside the isolated secure sandbox.
- Instantly review test case pass/fail indicators (5/5 Passed), execution latency (49.8ms), and algorithmic complexity rating.

Step 4: Virtual Audio Interview (Whisper Speech-to-Text)

- Review the screening prompt regarding technical problem-solving and architectural tradeoffs.
- Click '**Record Voice Answer**' to capture spoken response via your microphone, or submit existing audio.
- OpenAI Whisper converts the speech into structured text with live timestamps.
- Click '**Submit Voice Response & Evaluate**' to generate the STAR behavioral framework analysis (Situation, Task, Action, Result).

Step 5: Combined Merit Score & HR Shortlist Tracker

- Review the **Demographic-Blinded Fair Scoring Shield** ensuring all personal identifiers are masked.
- View the final aggregated **Merit-Based Score (97.3 / 100)** combining Skills, Code, and Interview performance.
- Follow the real-time application status timeline as your dossier progresses to the HR review queue.

Part 2: Recruiter Panel (8 Specialized Tabs Guide)

Tab 1: Executive Dashboard & Talent Funnel

Displays the recruitment pipeline funnel (Registered -> Parsed -> Screened -> Shortlisted), overall average scores, and active candidate roster cards with one-click full pipeline evaluation.

Tab 2: Candidate Profile & Smart Resume Extraction

Shows parsed candidate summary, PII anonymization status, extracted programming languages, databases, cloud tools, work history timeline, and education.

Tab 3: Role Competency & Skill Gap Analysis

Performs semantic similarity comparison between candidate skills and job descriptions. Displays compatibility gauges, matching strengths, and missing skill gap recommendations.

Tab 4: Live Sandboxed Coding Assessment

Allows recruiters to inspect candidate Python code, sandbox execution status, test case pass counts, execution runtime, and cyclomatic complexity ratings.

Tab 5: Virtual Screening & Speech-to-Text Analysis

Features direct audio file upload (.mp3, .wav, .m4a, .ogg) and mic recording. Displays the full transcribed speech text, timestamped segments, STAR breakdown, and communication clarity metrics.

Tab 6: Objective Demographic-Blind Scoring Matrix

Demonstrates the bias-prevention engine. Shows masked identity tags (CAND-001), objective weights (Skills 35%, Coding 35%, Interview 20%, Protocol 10%), and zero-bias compliance audit checks.

Tab 7: Comprehensive Candidate Evaluation Dossier

Consolidates all multi-agent scores into an executive briefing with an interactive competency radar chart and AI recommendation verdict.

Tab 8: Final Shortlisting & Automated HR Notification

Provides Human-in-the-Loop decision controls. Recruiters can Approve & Shortlist, Request Further Review, or Reject, and preview the automated personalized notification email dispatched to the candidate.